St. Francis Institute of Technology
(An Autonomous Institution)
AICTE Approved | Affiliated to University of Mumbai

A+ Grade by NAAC: CMPN, EXTC, INFT NBA Accredited: ISO 9001:2015 Certified

Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

Student Name: **Vishal Rajesh Mahajan** Student Roll No: **1**

Experiment – 3 : Infrastructure as A Service

Aim: To study and Implement Infrastructure As a Service using different cloud platform AWS / GC / AZURE (Free Tier Cloud Platforms).

Objective: After performing the experiment, the students will be able to –

- Launch an instance
- install any security updates
- execute bootstrap script
- EC2 instance store life
- access the instance metadata from the OS
- Amazon EBS Volume life

Lab objective mapped : ITL802 : To **implement** IAAS services.

Prerequisite: Concept of Operating System, Infrastucture.

Requirements: Cloud Login**,** Desktop, Browser, Internet etc.

Pre-Experiment Theory:

CLOUD SERVICES
IAAS:

IaaS stands for Infrastructure as a Service, a cloud computing model that provides on-demand access to virtualized computing resources like servers, storage, and networks over the internet, allowing businesses to rent IT infrastructure instead of buying and maintaining physical hardware. It's a core part of cloud services, offering flexibility and scalability, with the provider managing the underlying infrastructure and the user controlling operating systems, data, and applications.

Key aspects of IaaS:

- On-demand: Resources are available as needed, often on a pay-as-you-go basis.
- Virtualization: Delivers hardware (servers, storage, networks) in a virtualized, cloud-based environment.
- User responsibility: Users manage the OS, middleware, applications, and data.

- Provider responsibility: The provider handles the physical infrastructure (servers, storage, networking).

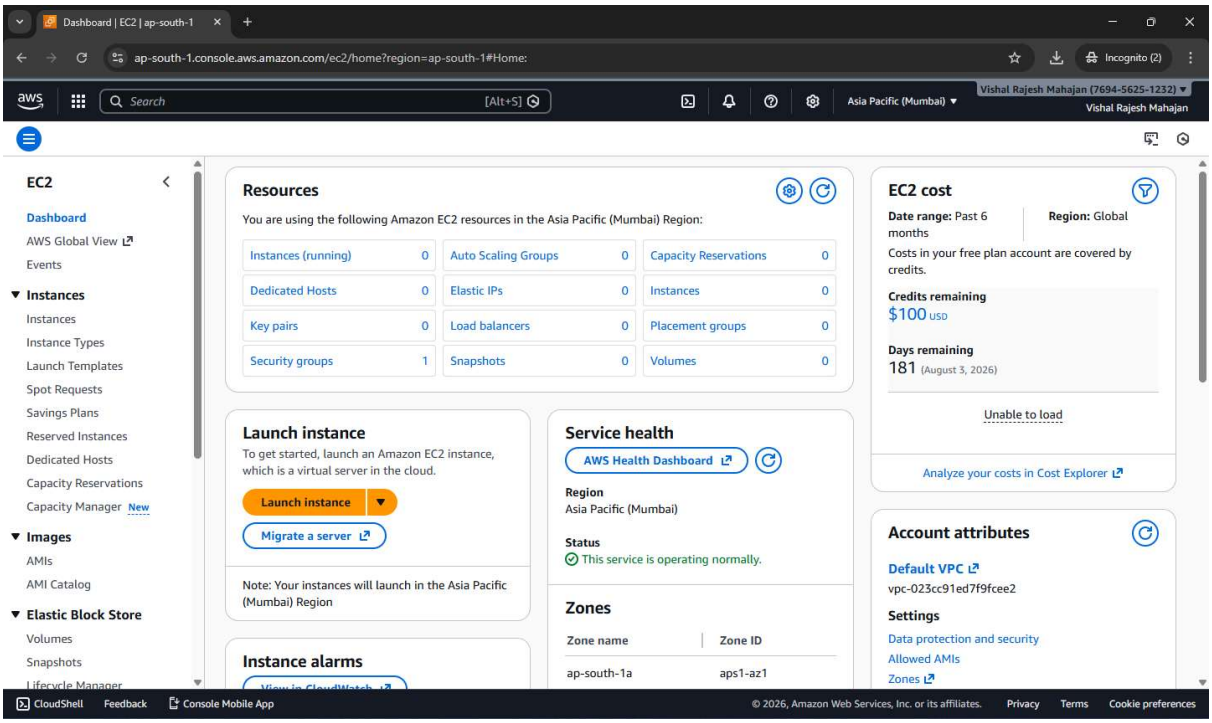
Examples of IaaS:

- Amazon Web Services (AWS) EC2, S3
- Microsoft Azure Virtual Machines, Storage
- Google Cloud Platform Compute Engine, Cloud Storage
- DigitalOcean, Rackspace

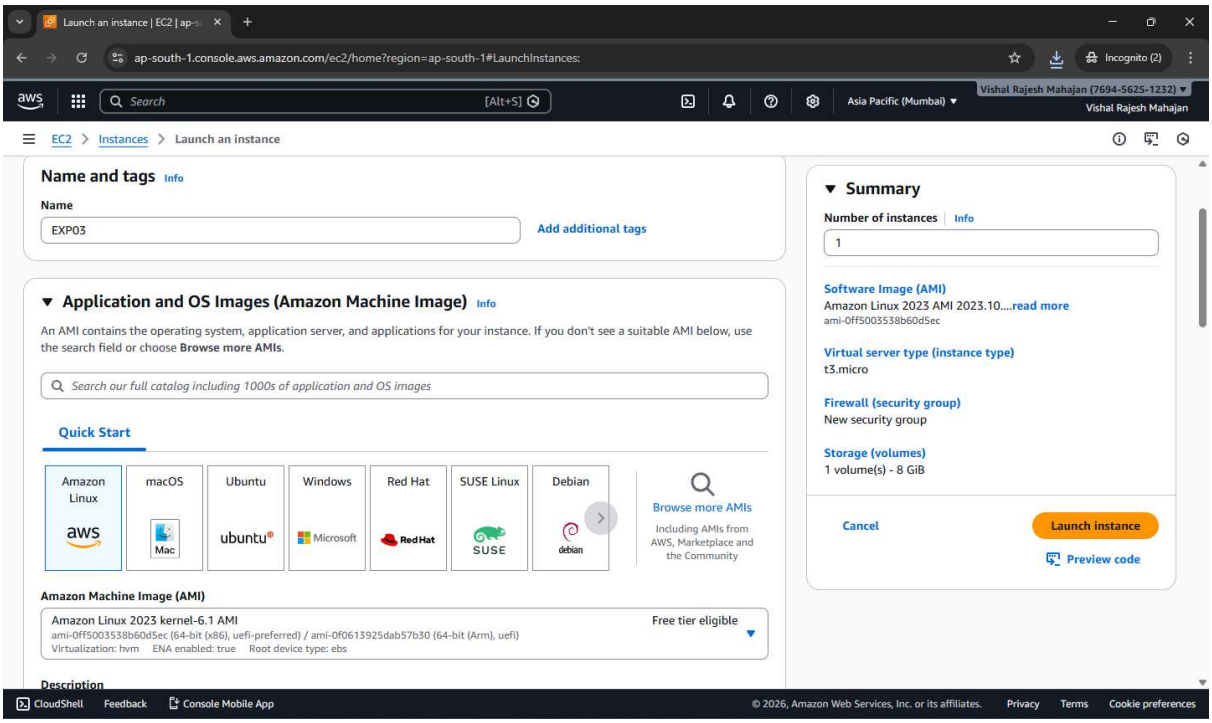
Procedure

Launch and Connect to a Linux Instance, log in with SSH, and install any security updates.

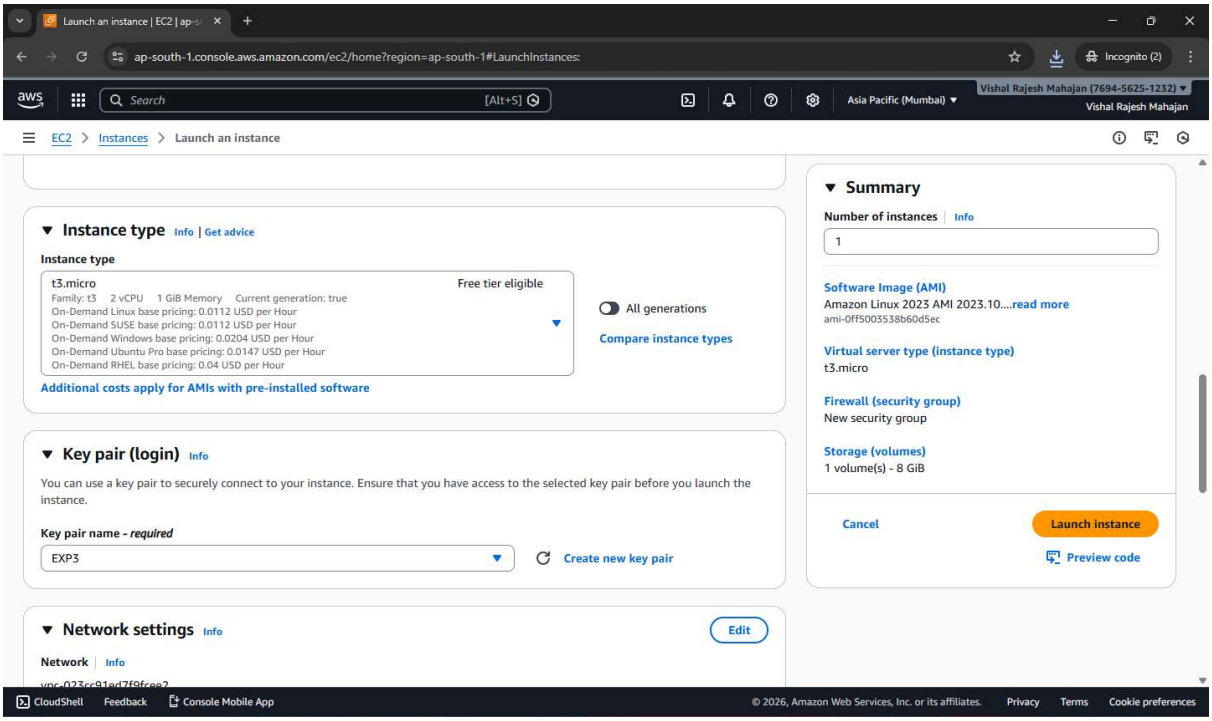
1. Launch an instance in the Amazon EC2 console.



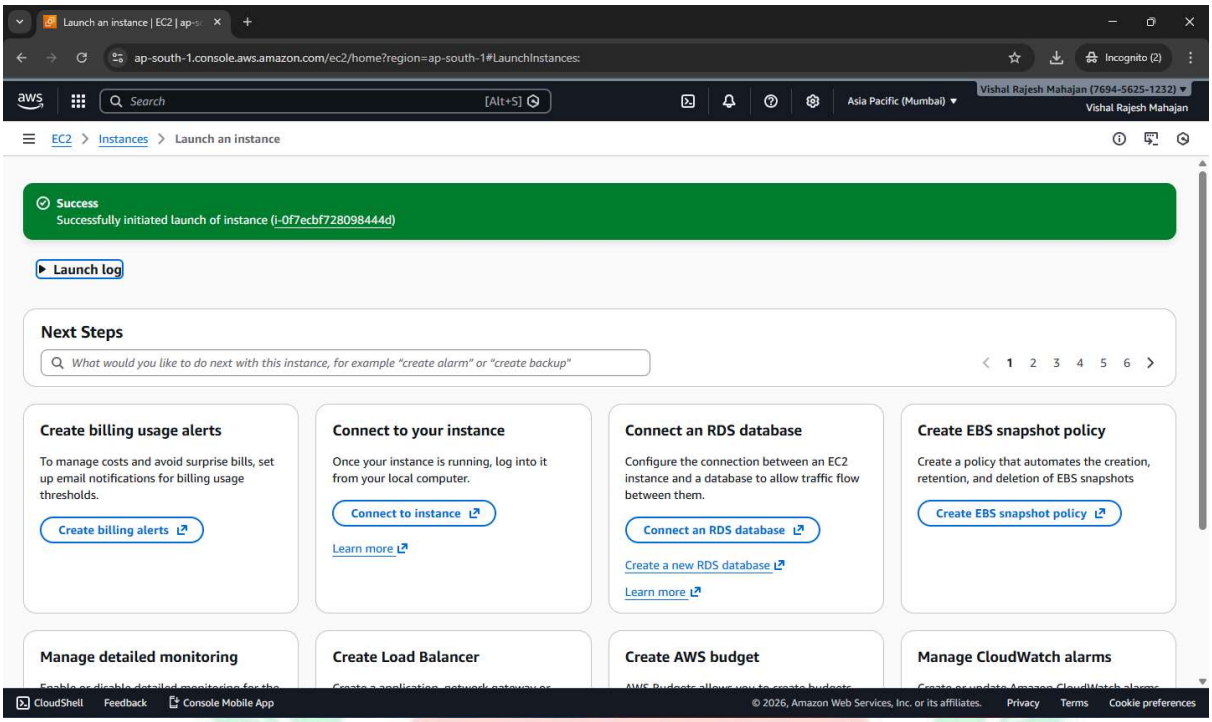
2. Choose the Amazon Linux AMI.



3. Choose the t3.micro instance type.



4. Launch the instance in the default VPC.



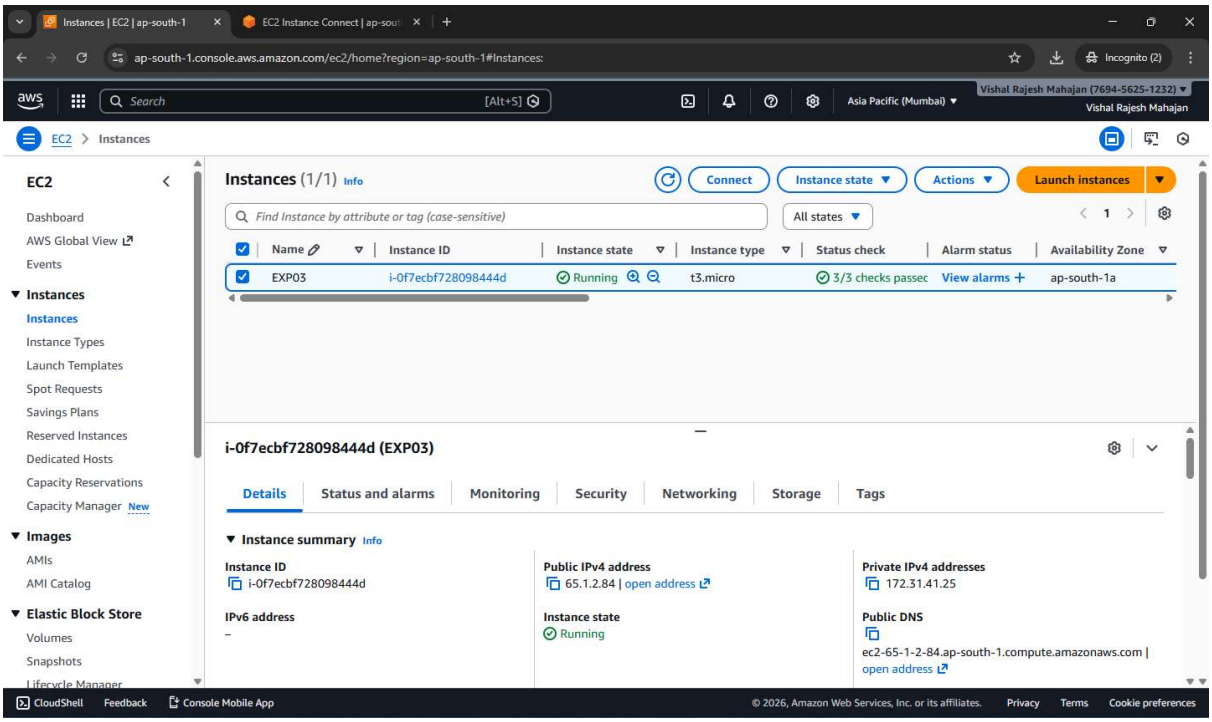
5. Assign the instance a public IP address.

6. Add a tag to the instance of Key: Name, Value: Exercise 3.1.

7. Create a new security group called Cert Book.

8. Add a rule to Cert Book allowing SSH access from the IP address of your (my IP) workstation (www.WhatsMyIP.org is a good way to determine your IP address).

9. Launch the instance.



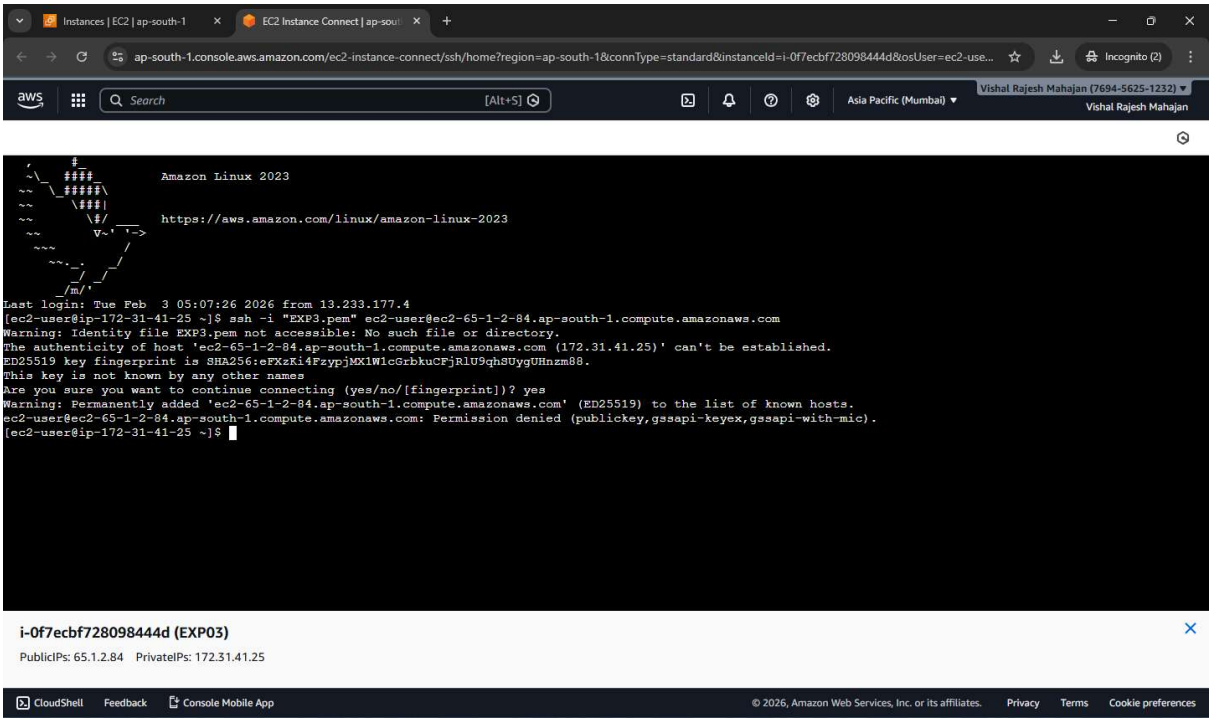
10. When prompted for a key pair, choose a key pair you already have or create a new

one and download the private portion.

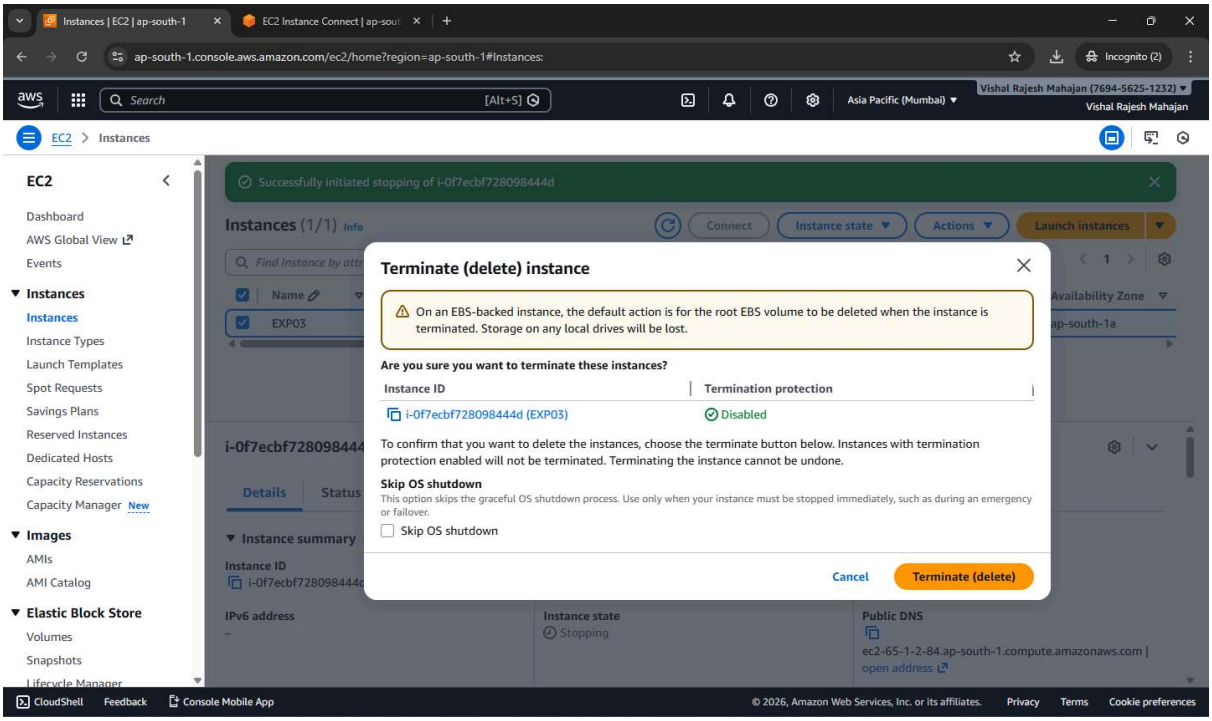
Amazon generates a keyname.pem file, and you will need a keyname.ppk file to connect to the instance via SSH. Puttygen.exe is one utility that will create a .ppk file from a .pem file.

11. SSH into the instance using the public IP address, the user name ec2-user, and the keyname.ppk file.

12. From the command-line prompt, run sudo yum update—security -y.



13. Close the SSH window.



Launch a Windows Instance with Bootstrapping

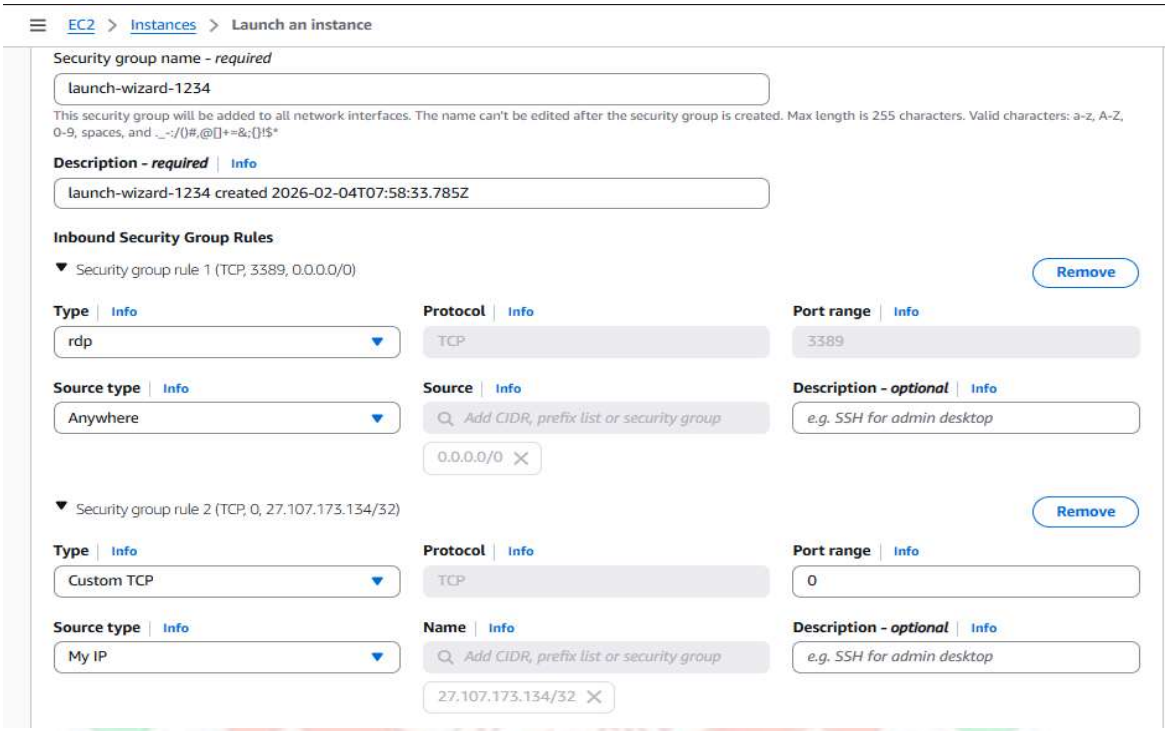
Specify a very simple bootstrap script. then confirm that the bootstrap script was executed on the instance.

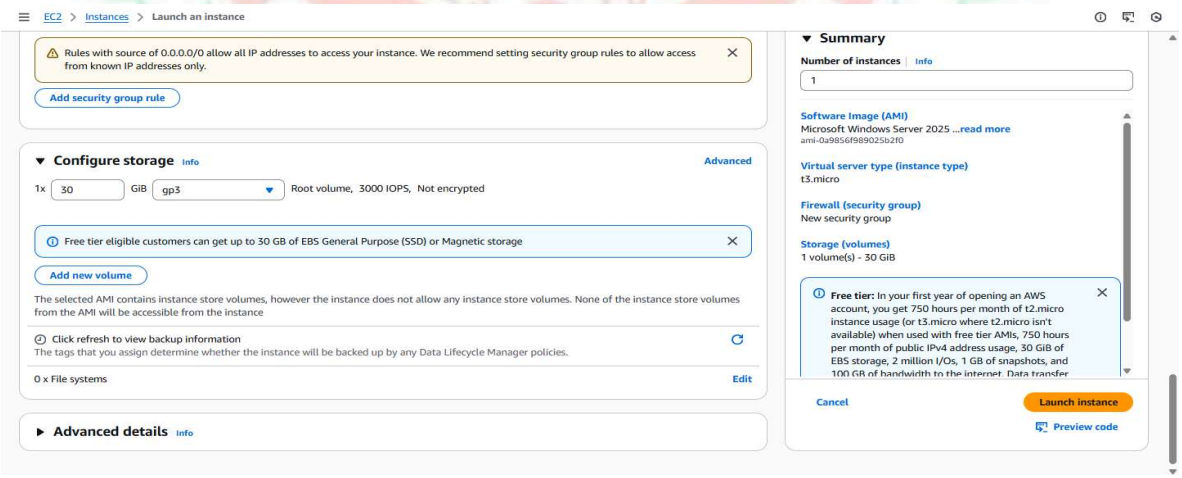
1. Launch an instance in the Amazon EC2 console.

2. Choose the Microsoft Windows Server 2012 Base AMI.

3. Choose the t3.micro instance type.

4. Launch the instance in either the default VPC.





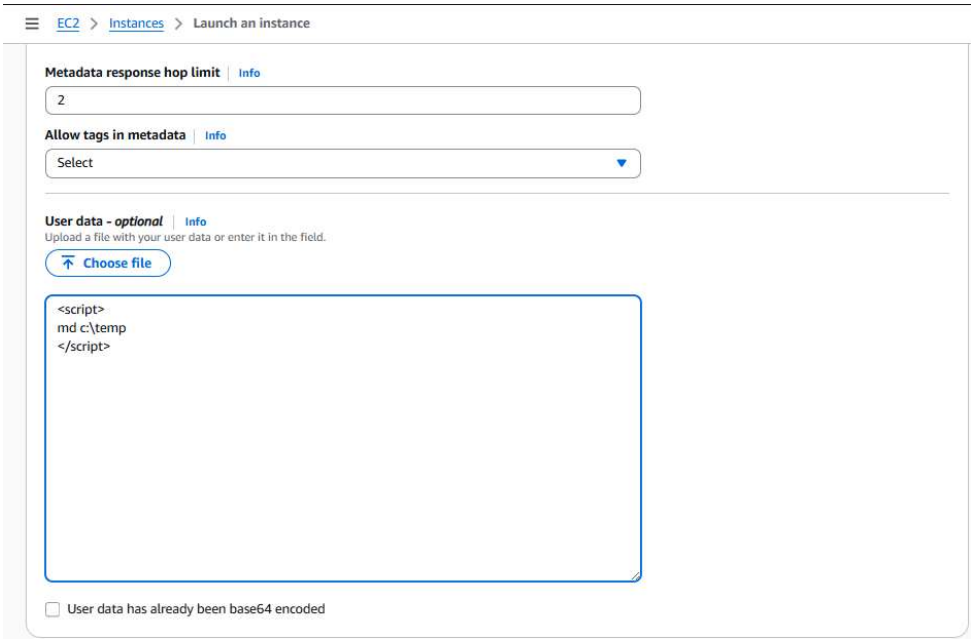
5. Assign the instance a public IP address.

6. In the Advanced Details section, enter the following text as UserData:

<script>

md c:\temp

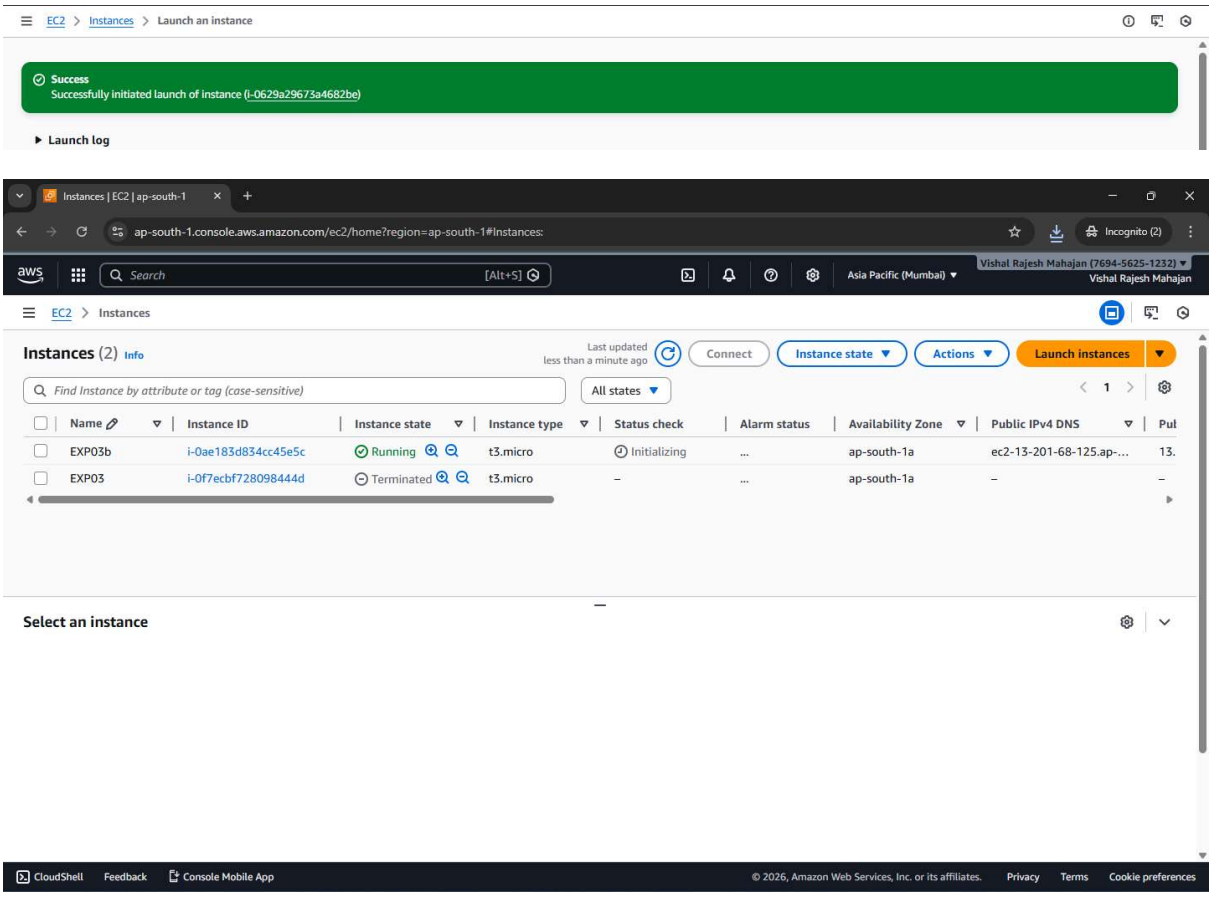
</script>



7. Add a tag to the instance of Key: Name, Value: Exercise 3.2.

8. Use the Cert Book security group from Exercise 3.1.

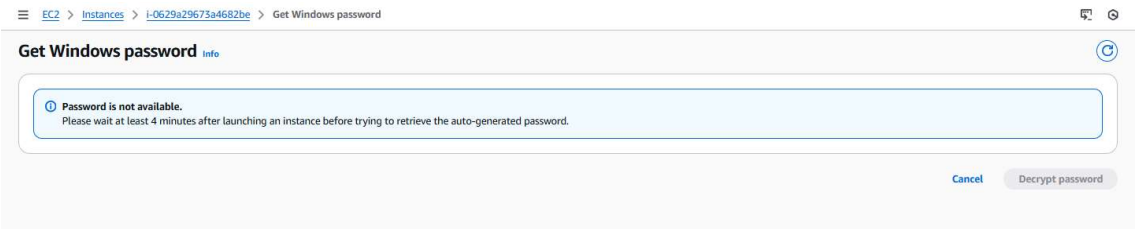
9. Launch the instance.

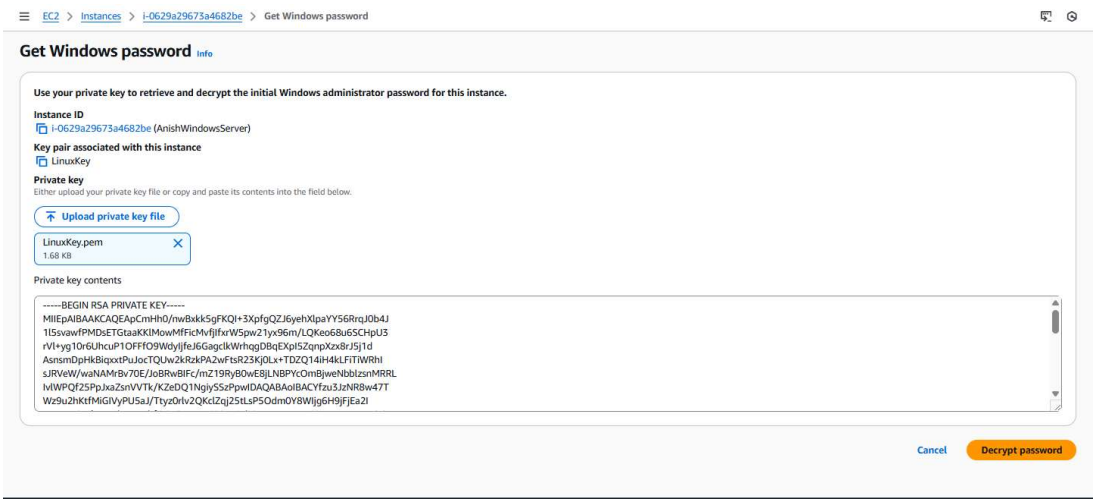


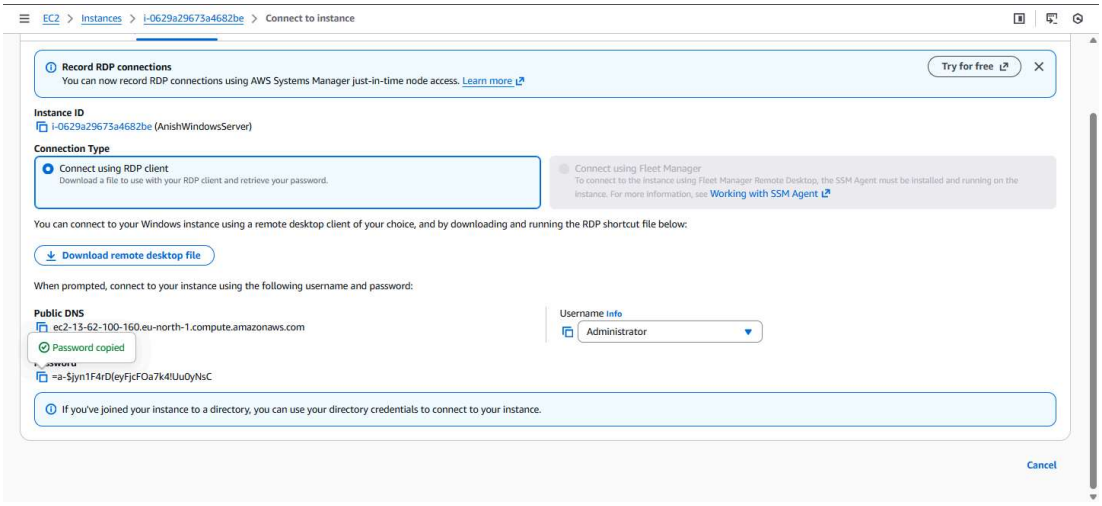
10. Use the key pair from Exercise 3.1.

11. On the Connect Instance UI, decrypt the administrator password and then download the RDP file to attempt to connect to the instance. Your attempt should fail because

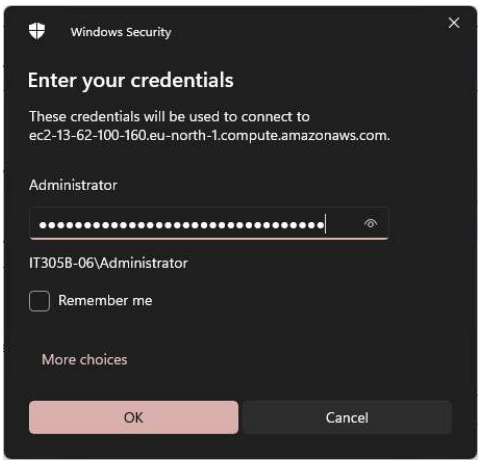
the Cert Book security group does not allow RDP access.







12. Open the Cert Book security group and add a rule that allows RDP access from your IP address.



13. Attempt to access the instance via RDP again.



14. Once the RDP session is connected, open Windows Explorer and confirm that the c:\temp folder has been created.

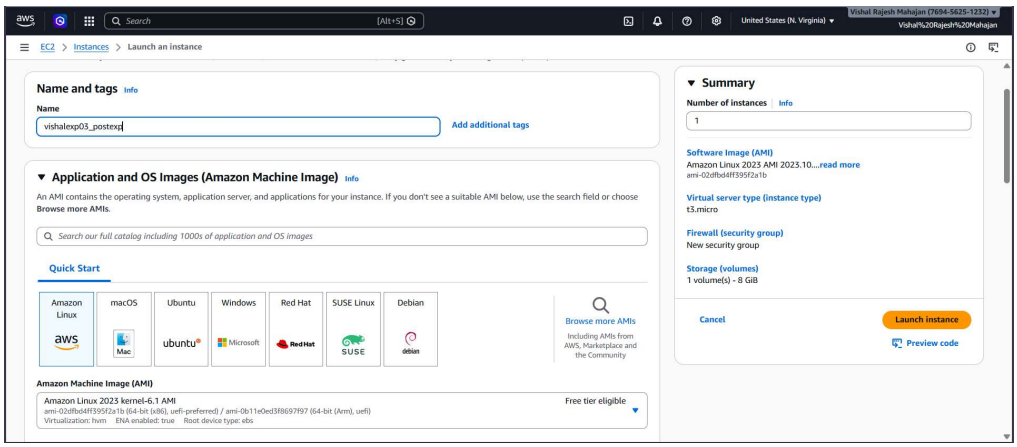
15. End the RDP session and terminate the instance.

Post-Experiments Exercise

Create an Amazon EBS Volume and observe that it remains after the instance is terminated. Amazon EBS volume persists beyond the life of an instance.

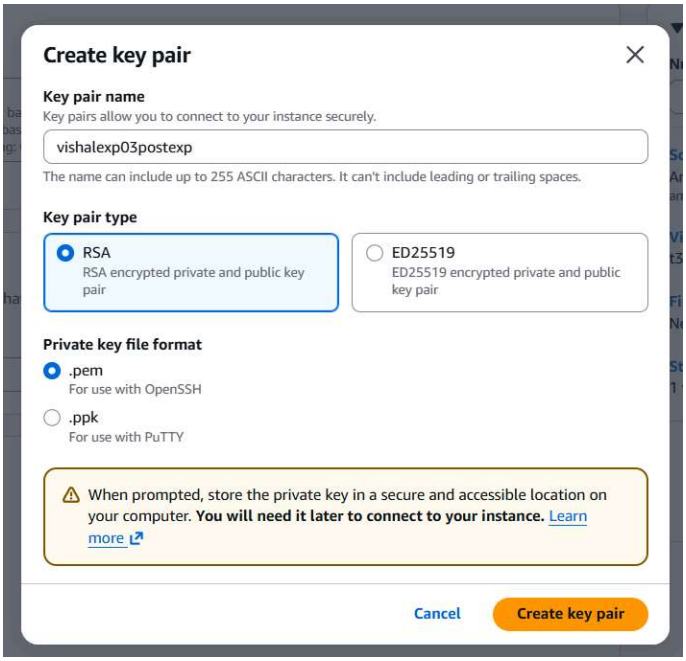
1. Launch an instance in the Amazon EC2 console.

2. Choose the Amazon Linux AMI.



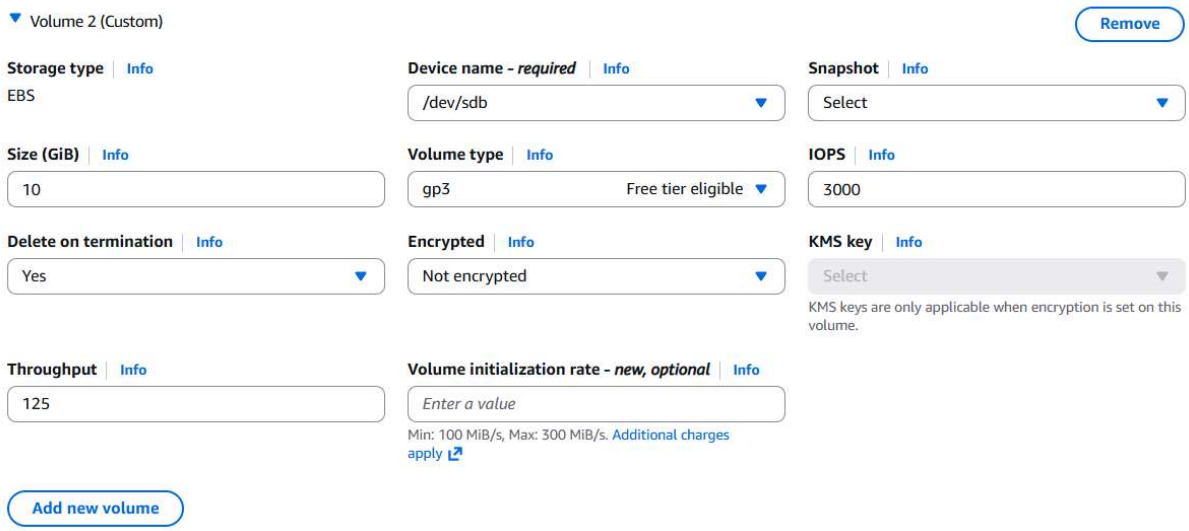
3. Choose the t3.micro instance type.





4. Launch the instance in the default VPC.

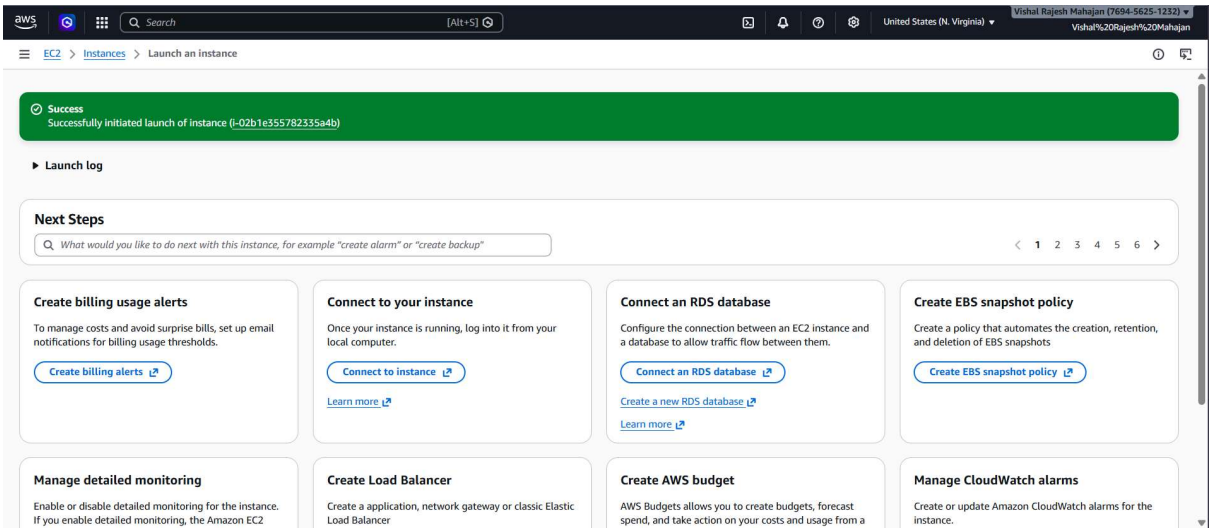
5. Add a second Amazon EBS volume of size 10 GB. Note that the Root Volume is set to Delete on Termination.



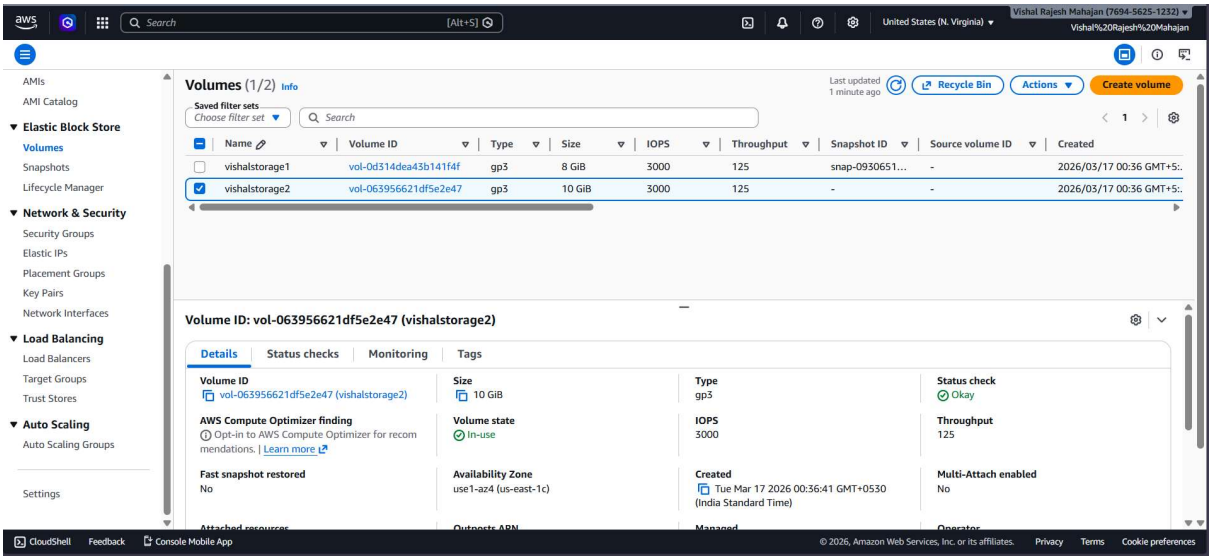
7. Add a tag to the instance of Key: Name, Value

8. Use the Cert Book security group from earlier exercises.

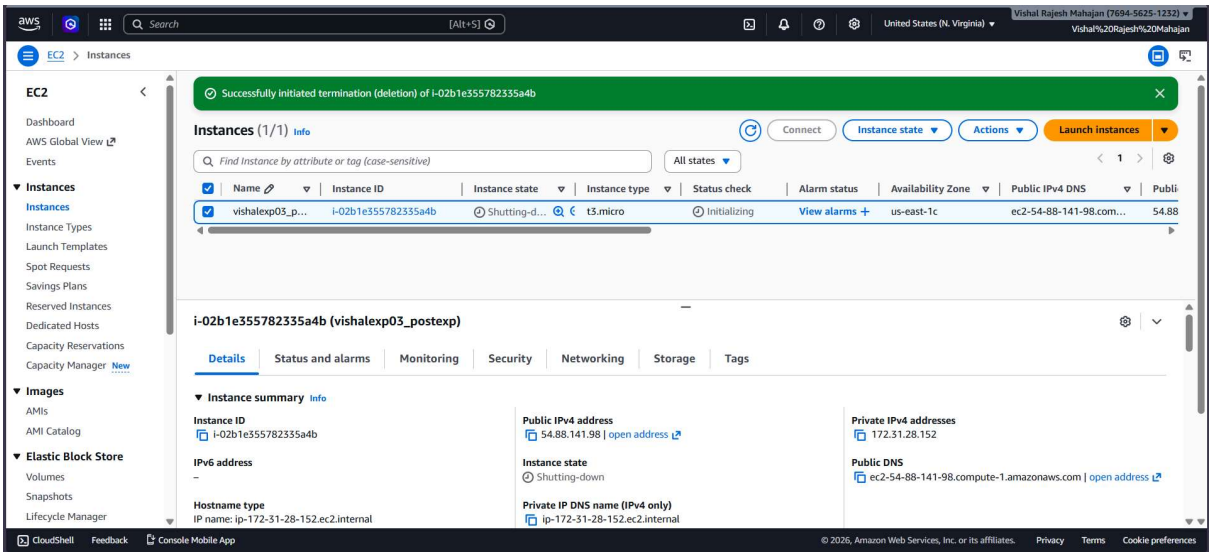
9. Launch the instance.

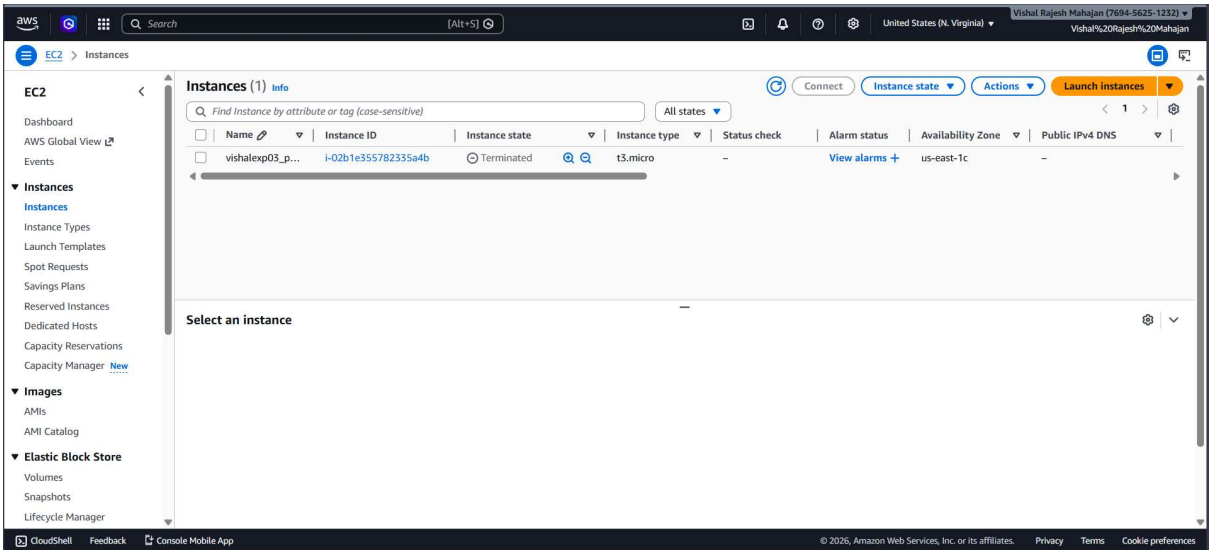


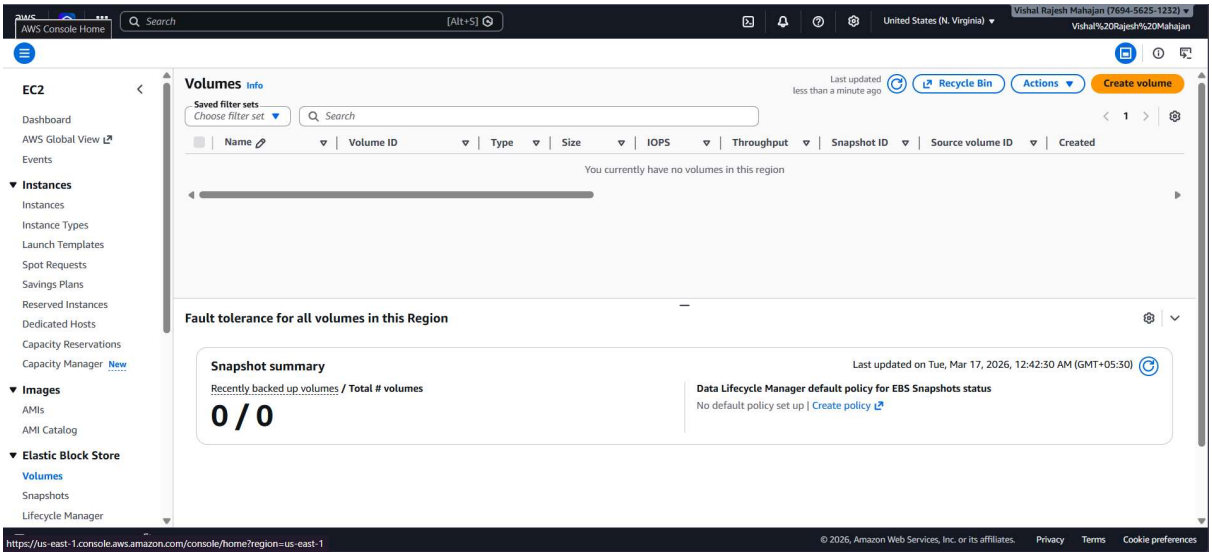
10. Find the two Amazon EBS volumes on the Amazon EBS console. Name them both



11. Terminate the instance.


Successfully initiated termination (deletion) of i-02b1e355782335a4b
Instances (1/1) Info
vishalexp03_p...
i-02b1e355782335a4b
Shutting-d...
t3.micro
Initializing
View alarms
us-east-1c
ec2-54-88-141-98.com...
i-02b1e355782335a4b (vishalexp03_postexp)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-02b1e355782335a4b
Public IPv4 address
54.88.141.98 | open address
Private IPv4 addresses
172.31.28.152
IPv6 address
Instance state
Shutting-down
Public DNS
ec2-54-88-141-98.compute-1.amazonaws.com | open address
Hostname type
IP name: ip-172-31-28-152.ec2.internal
Private IP DNS name (IPv4 only)
ip-172-31-28-152.ec2.internal


Instances (1) Info
vishalexp03_p...
i-02b1e355782335a4b
Terminated
t3.micro
View alarms
us-east-1c
Select an instance


Volumes Info
You currently have no volumes in this region
Fault tolerance for all volumes in this Region
Snapshot summary
Recently backed up volumes / Total # volumes
0 / 0
Data Lifecycle Manager default policy for EBS Snapshots status
No default policy set up | Create policy

Extended Theory:

1. What is bootstrap in a computer? (to be written in hand)
2. RDP (to be written in hand)
3. Discuss procedure to connect to your Linux instance using PuTTY (soft copy form)

To connect to a Linux instance via PuTTY from Windows, you must convert your .pem key file to .ppk format using PuTTYgen, open PuTTY, enter the instance IP address (Port 22, SSH), load your .ppk file under Connection > SSH > Auth > Credentials, and click "Open".

1. Step-by-Step Procedure
2. Convert Private Key (.pem to .ppk)
3. PuTTY requires a .ppk file, but many cloud providers provide a .pem file.
4. Launch PuTTYgen.
5. Click Load and select your .pem file.
6. Click Save private key to generate the .ppk file.
7. Configure PuTTY Session
8. Open PuTTY and go to the Session category.
9. In Host Name (or IP address), enter username@instance_ip (e.g., ec2-user@1.2.3.4).
10. Ensure Port is 22 and Connection type is SSH.
11. Load Private Key for Authentication
12. In the Category pane, expand Connection > SSH > Auth > Credentials (or just Auth in older versions).
13. Click Browse and select your .ppk file.
14. Connect to Instance
15. Click Open.
16. If a PuTTY Security Alert pops up, click Yes to trust the host.
17. A terminal window will open; if prompted, enter the username or password (if not using key-based authentication).
18. Common Connection Details
19. Default Users: ec2-user (Amazon Linux), ubuntu (Ubuntu), root (CentOS/Debian).
20. Port: 22 (Default SSH).

Results/Calculations/Observations:

Fill the following observation tables-(to be written in hand)

Sr. No		
1.	SSH port no	
2.	Linux VM Public IP	
3.	t2.micro RAM and CPU capacity	
4.	Decrypted pwd length	
5.	name of security grp for windows instance	

Post Experimental Exercise-
Questions:

1. Short note on EBS volume?(soft copy form)

Amazon Elastic Block Store (EBS) volumes are persistent, high-performance block-level storage devices designed for use with Amazon EC2 instances. Acting as virtual hard drives, they offer data persistence independent of the instance lifecycle, automatic replication within an Availability Zone for durability, and flexible, dynamic resizing capabilities.

Key Aspects of EBS Volumes:

- **Persistence & Flexibility:** Data remains intact even after stopping or terminating an EC2 instance. Volumes can be detached from one instance and attached to another.
- **Performance Types:** Available in SSD (General Purpose gp2/gp3, Provisioned IOPS io1/io2) for transactional workloads and HDD (Throughput Optimized st1, Cold sc1) for large-throughput workloads.
- **Durability & Backup:** Automatically replicated within an Availability Zone. Supports snapshots for backing up data to Amazon S3.
- **Scalability:** Sizes and performance (IOPS) can be modified on-the-fly without interrupting production.
- **Location Constraints:** EBS volumes are specific to an Availability Zone and cannot be attached to instances in different AZs

Conclusion:

1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:

[1] [Online] https://download.virtualbox.org/virtualbox/5.1.22/UserManual.pdf

[2] [Online] https://phoenixnap.com/kb/virtualbox-vs-vmware

[3] Samjhana Rayamajhi, Zinnia Sultana "Comparative Performance Analysis of the Virtualization Technologies in Cloud Computing", Volume 03, Issue 09 (September 2014),IJERT